# EF5560: FinTech and AI in Finance

Course Syllabus, Semester A 2026/27

City University of Hong Kong, Department of Economics and Finance

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| **Instructor** | Prof. Gavin Feng |
| **Email** | gavin.feng@cityu.edu.hk. Begin every subject line with `[EF5560]` so that your message is not lost among others |
| **Credit units** | 3 |
| **Medium** | English |
| **Prerequisites** | None. See *Course purpose* for what Classes 1–5 assume |
| **Class format** | Twelve three-hour seminars; exact dates, time, and venue are announced through the official timetable and Canvas |
| **Course site** | Official announcements, course materials, submission links, and assessment instructions will be posted on Canvas |
| **Consultation** | Zoom office hours at 10:00 on the Sundays listed on Canvas, where the link is also posted<br>Teaching assistants:<br>Yunting Liu, [contact removed]<br>Anna Zhao, [contact removed]<br>Copy me on any email you send them, with the same `[EF5560]` subject prefix |

## Course purpose

This course has two parts. In Classes 1–5, students use AI-assisted coding in R and Python to obtain financial data, construct predictors, study return predictability, compare linear and nonlinear models, interpret model outputs, and turn forecasts into portfolio decisions. You will not be asked to memorize code. You will be asked to define the problem, check when each variable became available, protect the validation and test samples, and report weak results as weak.

Classes 6–12 study how FinTech businesses work: the technology they use, how they make money, the risks they create, and how they are regulated. We cover blockchain, Bitcoin and Ethereum, decentralized finance, tokenization, stablecoins and central bank digital currencies, payment systems, lending, robo-advising, cybersecurity, RegTech, and selected AI applications for financial and compliance text.

Classes 1–5 build on each other, so regular attendance is important.

## Learning outcomes

The course aims to help students:

1. use selected AI-assisted R and Python workflows to organize, visualize, and check the timing of financial data;
2. interpret and compare selected regression and machine-learning methods for return prediction and portfolio applications using out-of-sample evidence, including the limits of coefficients, feature importance, and response profiles;
3. explain core mechanisms and business models in selected FinTech areas and recognize major financial, operational, cybersecurity, conduct, and regulatory risks; and
4. communicate an evidence-based analysis of a FinTech company or financial machine-learning application.

Coverage and depth depend on class progress. Students are assessed only on material taught in class or released through Canvas.

## Assessment

| Assessment component | Weight |
|---|---|
| FinTech company case report | 10% |
| Recorded video presentation | 20% |
| Machine-learning investment assignment | 20% |
| Participation and in-class work | 10% |
| Two-hour final examination | 40% |
| Total continuous assessment | 60% |
| Final examination | 40% |

**FinTech company case project.** Teams of up to three students submit a written case report (10%, no more than ten pages) and a recorded video presentation (20%, no more than ten minutes) in Week 12. Every member must deliver at least two minutes of substantive presentation. Submit one MP4 file no larger than 80 MB. There is no in-person presentation. A separate brief gives the framework, evidence standards, and rubric.

**Machine-learning investment assignment.** Teams of up to three submit one joint report, portfolio-weight CSV, and reproducible code package. It is released after the Week 5 class and is due two weeks later, in Week 7. The report cover lists all team members and CityU student numbers; one member is the designated submitter. The assignment brief gives the required analysis, submission instructions, and grading criteria.

**Teaching-assistant tutorials.** Two of those Sunday slots are run by a teaching assistant: 27 September 2026 on preparing the machine-learning assignment, and 29 November 2026 on preparing for the examination. Both are recorded.

**Participation.** Everyone starts at five marks out of ten and the mark moves during the semester.

- **Cold call.** I call on students by name rather than wait for volunteers, so expect to be asked.
- **A correct or well-reasoned answer earns one mark.** Sound reasoning counts even when the final answer is not the expected one.
- **A wrong answer costs nothing.** Being wrong, or passing, leaves the mark unchanged.
- **Being absent when called costs one mark.** Email me at least a day ahead if you cannot attend your registered class; otherwise the mark falls by one.

Questions, short exercises, and useful feedback to classmates count toward the same ten marks.

**Final examination.** The examination covers both course blocks and tests concepts, mechanisms, calculations, and critical comparison. It is handwritten and held during the examination period. A mock paper is released on Canvas in mid-November, and the second teaching-assistant tutorial works through it. Permitted materials will be confirmed on Canvas, subject to the final University examination instructions.

## Tentative teaching schedule

The order and emphasis may be adjusted according to class progress and current developments.

| Class | Block | Topic and focus |
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| 1 | ML in Finance | Financial data and vibe coding in R and Python |
| 2 | ML in Finance | Regression, return predictability, and out-of-sample design |

| Class | Block | Topic and focus |
|---|---|---|
| 3 | ML in Finance | Linear machine learning for return prediction |
| 4 | ML in Finance | Nonlinear machine learning for return prediction |
| 5 | ML in Finance | Interpretable machine learning and portfolio decisions |
| 6 | FinTech | FinTech overview and blockchain |
| 7 | FinTech | Bitcoin and Ethereum |
| 8 | FinTech | DeFi, NFTs, and real-world assets |
| 9 | FinTech | Stablecoins and CBDCs |
| 10 | FinTech | Payment systems |
| 11 | FinTech | Lending and crowdfunding |
| 12 | FinTech | Robo-advising, cybersecurity, RegTech, and AI applications to financial text, with a closing course synthesis |

## Course practices and policies

- **Evidence.** Time-sensitive claims about markets, companies, and regulation must be supported by a dated and credible source.
- **Reproducibility.** Empirical work must preserve the data timing and transformation steps needed to reproduce the reported result.
- **Generative AI.** AI-assisted coding is used in Classes 1–5. AI may be used in coursework when permitted by the relevant instructions and disclosed. Students remain responsible for all submitted work. Generative-AI tools are not permitted in the final examination.
- **Academic integrity.** Students must submit their own work, acknowledge others' words, graphics, ideas, and data, and report findings honestly. Plagiarism, fabrication, undisclosed copying, and submission of unverified generated material are serious offences under the University's Rules on Academic Honesty. All new students admitted in Semester A 2026/27 must complete the online tutorial and quiz and submit the declaration through the Academic Honesty website no later than **30 November 2026**.
- **Announcements.** Later Canvas or University announcements take precedence.

## Official course information

See the 2026/27 University course catalogue.